

Self-grading sheet, Question 1

Name _____

Date _____

A drug partly inhibits the Na⁺/K⁺ ATPase in a heart muscle cell. Trace the effect on intracellular Na⁺, on calcium removal by the Na⁺/Ca²⁺ exchanger, and on the strength of contraction.

- A. Intracellular Na⁺ falls, the exchanger removes more Ca²⁺, and contraction weakens
- B. Intracellular Ca²⁺ is unchanged, because the exchanger runs on ATP and does not depend on the pump
- C. Intracellular Na⁺ rises, the exchanger removes less Ca²⁺, and contraction strengthens
- D. Intracellular Na⁺ rises, the exchanger removes more Ca²⁺, and contraction weakens

The answer is C. The pump moves less Na⁺ out, so Na⁺ builds up inside and the Na⁺ gradient gets smaller. The Na⁺/Ca²⁺ exchanger is powered by that gradient, so it removes less Ca²⁺. More Ca²⁺ stays in the cell and is stored and released with each beat, so contraction is stronger. This is how digoxin works.

Be honest with yourself. A point counts only if you clearly drew it or said it. If you are not sure you did, it does not count.

Your model: check each point you clearly drew or labeled (6 points)

- ☐ The Na⁺/K⁺ ATPase: 3 Na⁺ out, 2 K⁺ in, powered by ATP.
- ☐ The Na⁺/Ca²⁺ exchanger: 3 Na⁺ in, 1 Ca²⁺ out, powered by the Na⁺ gradient, not by ATP (secondary active transport).
- ☐ The normal state: Na⁺ low inside and high outside, Ca²⁺ low inside.
- ☐ Partial inhibition: less Na⁺ pumped out, so intracellular Na⁺ rises and the Na⁺ gradient gets smaller.
- ☐ A smaller Na⁺ gradient: the exchanger removes less Ca²⁺, so more Ca²⁺ stays in the cell.
- ☐ More Ca²⁺: more force, so contraction strengthens.

Your teaching: check each point you clearly said (9 points)

- ☐ Read the question and said what it asks.
- ☐ Explained what the pump does and what powers it.
- ☐ Explained what the exchanger does and what powers it.
- ☐ Walked the chain in order, from the drug to the strength of contraction.
- ☐ A is wrong: inhibiting the pump makes Na⁺ rise inside, not fall.
- ☐ B is wrong: the exchanger does not use ATP; it depends on the Na⁺ gradient the pump makes.
- ☐ D is wrong: a smaller Na⁺ gradient means the exchanger removes less Ca²⁺, not more.
- ☐ Pointed to the spot on the model for each wrong choice.
- ☐ Gave the answer, C, and stated it again at the end.

Model points ____ of 6

Teaching points ____ of 9

Answer right? Yes / No

Your percent ____

Your score: add your model and teaching points, divide by 15, multiply by 75, and add 25 if your answer was right. Example: 5 model points and 7 teaching points is 12 of 15, which is 60; with the right answer, 85 percent.

No video? Grade your model from the photo of your whiteboard and leave the teaching points blank. Your score is then your model points divided by 6, times 75, plus 25 if your answer was right.

Self-grading sheet, Question 2

Name _____

Date _____

Ouabain is added to an intestinal epithelial cell. Trace the effect on intracellular Na⁺ and then on glucose uptake through SGLT at the apical membrane.

- A. Intracellular Na⁺ is unchanged, because SGLT keeps it low
- B. Intracellular Na⁺ rises, the Na⁺ gradient shrinks, and SGLT glucose uptake falls
- C. Intracellular Na⁺ rises, the Na⁺ gradient steepens, and SGLT glucose uptake rises
- D. Intracellular Na⁺ falls, and SGLT glucose uptake rises

The answer is B. Ouabain blocks the Na⁺/K⁺ ATPase on the basolateral membrane, so Na⁺ builds up inside the cell. The Na⁺ gradient across the apical membrane shrinks, and SGLT, which uses that gradient to pull glucose in uphill, takes up less glucose.

Be honest with yourself. A point counts only if you clearly drew it or said it. If you are not sure you did, it does not count.

Your model: check each point you clearly drew or labeled (6 points)

- ☐ The cell with the lumen on the apical side and the blood on the basolateral side, both labeled.
- ☐ SGLT on the apical membrane: Na⁺ and glucose in together, Na⁺ downhill and glucose uphill (secondary active transport).
- ☐ GLUT2 on the basolateral membrane: glucose out to the blood, downhill, by facilitated diffusion.
- ☐ The Na⁺/K⁺ ATPase on the basolateral membrane: 3 Na⁺ out, 2 K⁺ in, ATP, keeping Na⁺ low inside.
- ☐ Ouabain blocks the pump: intracellular Na⁺ rises and the Na⁺ gradient across the apical membrane shrinks.
- ☐ A smaller gradient: less energy for SGLT, so glucose uptake from the lumen falls.

Your teaching: check each point you clearly said (9 points)

- ☐ Read the question and said what it asks.
- ☐ Explained the apical and basolateral sides and which transporter sits on each.
- ☐ Explained what powers SGLT.
- ☐ Walked the chain in order, from ouabain to glucose uptake.
- ☐ A is wrong: SGLT brings Na⁺ in; it is the pump that keeps Na⁺ low.
- ☐ C is wrong: rising Na⁺ inside makes the gradient smaller, not steeper.
- ☐ D is wrong: blocking the pump makes Na⁺ rise inside, not fall.
- ☐ Pointed to the spot on the model for each wrong choice.
- ☐ Gave the answer, B, and stated it again at the end.

Model points ____ of 6

Teaching points ____ of 9

Answer right? Yes / No

Your percent ____

Your score: add your model and teaching points, divide by 15, multiply by 75, and add 25 if your answer was right. Example: 5 model points and 7 teaching points is 12 of 15, which is 60; with the right answer, 85 percent.

No video? Grade your model from the photo of your whiteboard and leave the teaching points blank. Your score is then your model points divided by 6, times 75, plus 25 if your answer was right.